

Siyu An

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Cloud-native Software Engineer specializing in Go-based distributed systems for DNS infrastructure, enterprise network security, and high-scale Kubernetes platforms across multi-cloud environments.

WORK EXPERIENCE

Infoblox Burnaby, BC • 08/2024 - Present

Software Engineer

- Engineered a high-throughput DNS logging pipeline in Go (DNSTAP + syslog over TCP/TLS) with WAL-backed buffering to handle backpressure at 10K QPS, reducing log loss from 1% to <0.3% and cutting router CPU usage by 42%.
- Orchestrated a CloudQuery-powered multi-cloud DNS state ingestion fabric integrating Cloudflare and Akamai to unify configuration topology across hybrid environments, operationalizing 200+ enterprise networks and driving a 22% uplift in feature adoption.
- Extended BloxOne DDI by implementing RPZ-based DNS response filtering and Stub Zone support in Kubernetes-native Go services, improving security enforcement and zone resolution consistency across distributed deployments.
- Introduced an MCP-powered CI/CD orchestration system that enables safe AI-assisted deployments through structured runbooks and guardrails, reducing manual release effort by 80% and improving cross-team release consistency.
- Implemented a Go-based static analysis tool to analyze multi-repository dependency graphs and detect inconsistent imports across services, reducing cross-service review overhead and improving dependency alignment.

SFU Big Data Hub Burnaby, BC • 05/2023 - 09/2023

Data Engineer Co-op

- Co-authored a research publication in Nature Scientific Data, contributing to large-scale data modeling and analytics for EV charging infrastructure planning.
- Designed and implemented ETL pipelines using Python and Pandas to process 200M+ records, reducing end-to-end data processing latency by 25%.
- Developed a Web API client to automate external data synchronization workflows, reducing manual data acquisition effort by 40%.

Best Energy Equipment Tianjin, China • 03/2019 - 10/2019

Full Stack Engineer • Full-time

- Developed backend services for an IoT-based pipeline monitoring system using Java, Spring Boot, MQTT, and REST APIs to support real-time device data ingestion and web dashboard integration.
- Implemented and containerized backend modules with Docker, supporting system integration between industrial devices and web applications and improving deployment consistency across environments.

SKILLS

Languages: Go, Java, Python, SQL, TypeScript

Cloud Infrastructure: AWS, Docker, GCP, Grafana, Kubernetes, Linux, Prometheus, Terraform

Distributed Systems & Networking: Akamai, Cloudflare, DNS, etcd, gRPC, Kafka, Microservices, REST APIs, RPZ

EDUCATION

MSc in Computer Science

Simon Fraser University

Burnaby • 09/2022 - 01/2024

MSc in Data Science and Analytics

University of Leeds

Leeds • 09/2020 - 11/2021

BEng in Materials Science and Engineering

Hebei University of Architecture

China • 09/2013 - 07/2017